■ VoiceShield AI: Comprehensive Technical Documentation

Smart India Hackathon (SIH 2026)

Problem Statement ID: 26104

Title: AI-Powered Real-Time Detection and Prevention of Voice Cloning Impersonation Attacks

Theme: Blockchain & Cybersecurity

Nodal Ministry / Department: Cyber Security Cell, AICTE

Table of Contents

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1. Executive Summary & Problem Statement

1.1 The Threat Landscape

Recent breakthroughs in generative artificial intelligence, neural vocoders (HiFi-GAN, WaveGlow, MelGAN), and diffusion-based speech synthesis (DiffWave, VITS) have enabled threat actors to clone human voices with photographic fidelity using less than 3 seconds of reference audio.

These generative clones are actively weaponized in:

- **High-Value Executive / CXO Fraud**: Impersonating Chief Executive Officers or Chief Financial Officers over VoIP/telephony to instruct urgent multi-crore bank wire transfers.
- **Government Official Impersonation**: Spoofing high-ranking civil or military authorities to extract restricted intelligence.
- **Social Engineering & OTP Interception**: Tricking bank call-center agents or telecom operators into resetting multi-factor authentication (MFA) or transferring SIM identities.

1.2 The Core Problem with Conventional Deepfake Detectors

Most open-source or commercial deepfake voice detectors rely on a **single monolithic neural network** trained on a single synthetic speech corpus. As proven during real-world acoustic replay and vocoder transfer, single-model systems suffer catastrophic generalization failure when encountering:

- Codec transcoding artifacts (e.g., Opus compression over WhatsApp or VoIP).
- Acoustic room impulse responses (replaying speech through a loudspeaker in an office).
- Unseen neural vocoder architectures.

1.3 VoiceShield AI Solution

VoiceShield AI solves this generalization gap through a **5-layer defense-in-depth matrix** combining physical vocal tract inverse filtering, spectral group-delay phase continuity, biological micro-prosodic tracking, speaker acoustic centroid verification, and a custom **Conformer neural network with Attentive Statistics Pooling (ASP)** trained from scratch.

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2. Datasets & Data Processing Pipeline

2.1 Dataset Ingestion & Composition

The system was trained from scratch using two balanced corpora to ensure resistance against out-of-domain transfer:

Dataset Corpus	Origin & Format	Content Description	Processed Samples
Kaggle DEEP-VOICE Benchmark	`birdy654/deep-voice-deepfake-voice-recognition`	42 multi-minute audio files: 21 genuine human speech files (`Real.zip`) and 21 neural clone files (`Fake.zip`).	**1,470 balanced samples** (735 Real, 735 Fake)
Diverse Vocoders & Replay Corpus	Custom Forensic Benchmark (`data/cached_features.npz`)	Synthetic vocoder samples (HiFi-GAN, WaveGlow, robotic pitch flatness) and room impulse response replay attacks.	**300 balanced samples** (150 Real, 150 Fake)
Master Training Dataset	Unified & Stratified Master Corpus	Consolidated multi-source benchmark.	**1,770 total samples** (885 Real, 885 Fake)

2.2 Audio Preprocessing Pipeline

1. **Universal Codec Resampling**: All audio (WAV, MP3, OGG, WebM, FLAC, M4A) is ingested and standardized to **16,000 Hz, 1-channel Mono, 32-bit Floating Point PCM** via FFmpeg and SoundFile.
2. **DC Offset Removal & Peak Normalization**:

$$\tilde{x}[n] = \frac{x[n] - \mu_x}{\max(|x[n] - \mu_x|) + \epsilon}$$

Prevents dynamic range clipping and normalizes gain across microphones.

3. **Voice Activity Detection (VAD)**:

A 30ms sliding window energy filter identifies and strips unvoiced silence, preventing silent background frames from diluting speech statistics.

2.3 Feature Matrix Extraction

Each audio segment is transformed into a standardized **100×32 spatio-temporal matrix**:

- **13 Mel-Frequency Cepstral Coefficients (MFCCs)**: Encodes low-to-mid frequency vocal tract resonances.

- **13 Linear-Frequency Cepstral Coefficients (LFCCs)**: Linear filterbank spacing capturing critical high-frequency harmonic boundaries ($>4\text{ kHz}$) where neural vocoders produce subtle artifacts.
- **6 Vocoder Spectral Descriptors**:

1. *Spectral Centroid* (brightness / center of spectral mass)
2. *Spectral Rolloff 85%*
3. *Spectral Rolloff 95%*
4. *Spectral Flatness* (geometric mean / arithmetic mean of power spectrum)
5. *High-Frequency Energy Ratio* ($\sum_{f \ge 4000\text{Hz}} |X(f)|^2 / \sum |X(f)|^2$)
6. *Spectral Flux* (spectral rate of change between adjacent frames)

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3. End-to-End System Workflow

```
                              INCOMING CALL / INTERCEPT
                         (Web Browser Mic, VoIP Stream, File)
                                           ■
                                           ▼
                         AUDIO INGESTION & PRIVACY ENCLAVE
                 16kHz Mono • VAD Silence Stripping • Ephemeral RAM
                                           ■
      ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
      ■                    ■                               ■                   ■
      ▼                    ▼                               ▼                   ▼
  VECTOR 1:           VECTOR 2:                       VECTOR 3:           VECTOR 4:
Deep Conformer    LPC Glottal Flow                Phase-Aware MGD     Biometric Micro-
Neural Backbone   Inverse Residual                Group Delay         Prosody Tracking
(Conv1D + ASP)    (Residual Kurtosis)             (Phase Dispersion)  (F0 Jitter/Shimmer)
      ■                    ■                               ■                   ■
      ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
                           ■
                           ▼
                      VECTOR 5:
             Speaker Identity Verification
           (Cosine distance against CXO profile)
                           ■
                           ▼
            DYNAMIC MULTI-SIGNAL RISK ENGINE
         (Context Multipliers: ■ Amount, CXO Flag, OTP)
                           ■
      ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
      ▼                    ▼                    ▼
■ LOW RISK           ■ SUSPICIOUS        ■ HIGH RISK
 (< 35% Score)        (35% - 70% Score)   (> 70% Score)
Authentic Voice       Secondary Verify    Active Deepfake Attack
      ■                    ■                    ■
      ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
                           ■
                           ▼
            AUTOMATED PREVENTION PLAYBOOK
      • API Hard Freeze on Wire Transfers / Transaction Block
      • Out-of-Band Callback via Pre-Registered Telecom Trunk
      • Step-Up Hardware Token / Biometric Verification
                           ■
                           ▼
             ENTERPRISE SIEM & SOC TELEMETRY
      • MITRE ATT&CK T1656 / T1656.001 Mapping
      • RFC-5424 Common Event Format (CEF) Syslog Generation
      • India DPDP Act 2023 & GDPR Cryptographic Zero-Retention
```

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4. 5-Vector Multi-Layer Forensic Architecture

Vector 1: Multi-Scale Conformer Deep Neural Net

- **Location**: `core/conformer_model.py`
- **Principle**: Combines Depthwise Separable 1D Convolutions (capturing micro-temporal vocoder glitches) with Multi-Head Self-Attention (MHSA) (modeling macro-temporal speech rhythm) and Attentive Statistics Pooling (ASP).

Vector 2: LPC Glottal Flow Inverse Residual Analysis

- **Location**: `core/glottal_forensics.py`
- **Physiological Principle**: In human biology, voiced speech is generated by non-linear acoustic shockwaves produced as the vocal folds snap shut during glottal closure. This excitation signal $e[n]$ is filtered by the vocal tract filter $V(z)$.
- **Mathematical Inversion**:

Using 16th-order Levinson-Durbin Linear Predictive Coding (LPC), we estimate vocal tract coefficients a_k and isolate the glottal flow residual:

$$e[n] = x[n] - \sum_{k=1}^{16} a_k x[n-k]$$

- **Forensic Detection**:
- Human glottal residuals exhibit high non-Gaussian impulsivity with **Kurtosis ≥ 15.0**.
- Neural vocoders produce over-smoothed Gaussian noise residuals (**Kurtosis < 4.2**) or non-physical mathematical boundary clicks (**Kurtosis > 110.0**).

Vector 3: Phase-Aware Modified Group Delay (MGD)

- **Location**: `core/spectral_phase.py`
- **Principle**: Standard neural vocoders (e.g. HiFi-GAN, WaveGlow) synthesize audio using Inverse Short-Time Fourier Transforms (iSTFT) or learned upsampling convolutions. While magnitude spectra are well-reconstructed, the phase angles $\phi(f)$ contain subtle phase dispersion, high-frequency phase derivative jitter, and phase wrap tears.
- **Metrics**:
- Modified Group Delay (MGD) across sub-bands.
- High-frequency ($>4\text{ kHz}$) phase 2nd-derivative standard deviation ($\nabla^2 \phi(f)$).

Vector 4: Biometric Micro-Prosody Tracking

- **Location**: `core/feature_extraction.py`
- **Principle**: Human vocal cords cannot sustain mathematically invariant frequencies. Involuntary micro-perturbations occur continuously:
- **Jitter**: Cycle-to-cycle variation in fundamental frequency F_0.
- **Shimmer**: Cycle-to-cycle variation in speech amplitude.
- **Pitch Std (F_0)**: Synthetic clones often sound robotic (near 0 Hz pitch standard deviation) or drift unnaturally across unvoiced phonemes.

Vector 5: Speaker Acoustic Centroid Verification

- **Location**: `core/speaker_verifier.py`

- **Principle**: Computes a 78-dimensional acoustic centroid voiceprint embedding from enrolled executive/CXO audio. During a live call, if the caller claims to be the CEO, cosine similarity is measured:

$$\text{Similarity} = \frac{\mathbf{v}_{\text{claimed}} \cdot \mathbf{v}_{\text{incoming}}}{\|\mathbf{v}_{\text{claimed}}\| \|\mathbf{v}_{\text{incoming}}\|}$$

A mismatch immediately flags an active **impersonation attack** regardless of whether the voice is real or synthetic.

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5. Deep Learning Model: Enterprise Voice Conformer

5.1 Architecture Overview

The model (**`EnterpriseVoiceConformer`**) consists of:

1. **Input Projection**: Linear layer projecting 32-dimensional features to $d_{\text{model}} = 64$ with LayerNorm and Dropout (0.1).
2. **Macaron-Style Conformer Blocks (2 Layers)**:

- Half-Step Feed-Forward Module (FFN 1 with GELU activation).
- Multi-Head Self-Attention Module (4 attention heads, $d_{\text{k}} = 16$).
- Depthwise Separable 1D Convolution Module (Kernel size = 7, BatchNorm1d, GLU gating).
- Half-Step Feed-Forward Module (FFN 2).
- Final LayerNorm.

3. **Attentive Statistics Pooling (ASP)**:

Instead of simple global average pooling, ASP computes attention weights α_t over all frames:

$$\mu = \sum_{t=1}^T \alpha_t \mathbf{h}_t, \quad \sigma = \sqrt{\sum_{t=1}^T \alpha_t (\mathbf{h}_t - \mu)^2}$$

Concatenating $[\mu; \sigma]$ yields a 128-dimensional pooled representation that captures both the mean acoustic profile and temporal variation.

4. **Classification MLP Head**: Linear(128, 64) $\to$ LayerNorm $\to$ GELU $\to$ Dropout(0.25) $\to$ Linear(64, 2).

5.2 Training Hyperparameters

- **Optimizer**: AdamW ($\beta_1=0.9, \beta_2=0.999$, weight decay = 10^{-4})
- **Learning Rate Scheduler**: Cosine Annealing ($T_{\max} = 25$ epochs, initial LR = 10^{-3})
- **Loss Function**: Cross-Entropy Loss with gradient clipping ($\|\mathbf{g}\| \le 1.0$)
- **Batch Size**: 32
- **Hardware Profile**: CPU-optimized (trainable on CPU in ~45 seconds) with CUDA acceleration enabled on NVIDIA GPUs (RTX 4060).

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6. Dynamic Risk Engine & Automated Prevention Playbooks

6.1 Multi-Vector Risk Formulation

The risk engine synthesizes all 5 vectors into a composite score:

$$\text{Base Risk} = w_1 \cdot P_{\text{neural}} + w_2 \cdot S_{\text{spectral}} + w_3 \cdot S_{\text{prosody}} + w_4 \cdot S_{\text{glottal}} + w_5 \cdot S_{\text{phase}} + w_6 \cdot S_{\text{speaker}}$$

Where weights are calibrated to:

- $w_1 = 0.35$ (Conformer Neural Model)
- $w_2 = 0.18$ (Vocoder Spectral Descriptors)
- $w_3 = 0.15$ (Prosody / Pitch Contour)
- $w_4 = 0.16$ (LPC Glottal Flow Residual Kurtosis)
- $w_5 = 0.16$ (Phase MGD Coherence)
- $w_6 = 0.10$ (Speaker Verification Cosine Distance)

6.2 Contextual Enterprise Multipliers

In real banking and enterprise environments, risk increases with financial exposure:

- **High-Value Transaction**: $> \$10,000$ (or ■8,00,000) adds $+0.10$ to $+0.25$ risk.
- **Executive / CXO Target Call**: Adds $+0.12$ risk.
- **Sensitive Credential / OTP Request**: Adds $+0.15$ risk.

6.3 Automated Prevention Playbooks

Based on composite risk, the system executes real-time countermeasures:

Risk Tier	Composite Score	Visual Badge	Automated Defense Playbook
LOW	$0.00 - 0.34$	■ LOW RISK	Allow call; continuous passive monitoring; standard logging.
SUSPICIOUS	$0.35 - 0.69$	■ SUSPICIOUS	**Step-Up Verification**: Suspend unverified wire transfer approvals; initiate secondary SMS/App OTP verification; flag call in SOC console.
HIGH (ATTACK)	$0.70 - 1.00$	■ HIGH RISK	**Immediate Freeze**: Hard block financial transaction API; terminate authorization session; trigger automated out-of-band callback to pre-registered trunk; dispatch high-severity SIEM alert.

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7. Real-Time Sliding Window Streaming Engine

For live telephony and VoIP collaboration platforms (Zoom, Teams, SIP trunks), VoiceShield AI implements a **3-second sliding window stream processor** (`core/stream_detector.py`):

1. **Consecutive Anomaly Tracking**: A single noisy audio packet does not trigger a false alarm. The system requires consecutive anomalies across sliding windows to escalate threat levels.
2. **Exponential Moving Average (EMA)**:

$$R_{\text{rolling}}[t] = \alpha \cdot R_{\text{chunk}}[t] + (1 - \alpha) \cdot R_{\text{rolling}}[t-1] \quad (\alpha = 0.45)$$

3. **Continuous Audio Container Encapsulation**: In the frontend (`ui/app.js`), the MediaRecorder cycles every 3 seconds to ensure every transmitted chunk has complete EBML/WAV headers, avoiding codec parsing drops.

8. Enterprise SIEM Telemetry & Regulatory Privacy Compliance

8.1 MITRE ATT&CK Mapping

Every detection event is mapped to industry threat frameworks:

- **Tactic**: `TA0001` (Initial Access / Social Engineering)
- **Technique**: `T1656` (Impersonation)
- **Sub-Technique**: `T1656.001` (AI Voice Cloning / Audio Deepfake Impersonation)

8.2 Common Event Format (CEF) & RFC-5424 Syslog

The platform automatically generates production-ready CEF events for enterprise SIEM ingestion (Splunk, Microsoft Sentinel, IBM QRadar, CrowdStrike Falcon):

```
CEF:0|VoiceShieldAI|VoiceDefensePlatform|2.0|HIGH|Voice Clone Impersonation Detected|10|incidentId=INC-9635F7E8 callId=LIVE-7
```

8.3 Privacy Enclave: India DPDP Act 2023 & GDPR Compliance

- **Zero-Retention Ephemeral RAM**: Audio waveforms exist only in volatile memory during feature extraction and are wiped immediately via `PrivacyComplianceGuard.purge_raw_audio(waveform)` (zero-filling memory arrays).
- **Cryptographic Non-Reversible Telemetry**: Only a one-way **SHA-256 audio fingerprint** is recorded in audit logs, making it impossible to reconstruct raw voice recordings from audit logs.

9. Universal Transferable Model Formats (ONNX & TorchScript)

To integrate seamlessly with external models, partner systems, or other team members' laptops without requiring this full codebase, the model has been exported to the `model_export/` folder:

Asset	Format & Size	Portability Advantages
`voiceshield_conformer.onnx`	**Universal ONNX (625 KB)**	Framework-agnostic. Runs with `pip install onnxruntime`. **No PyTorch needed.**
`voiceshield_conformer.pt`	**TorchScript (670 KB)**	Self-contained traced PyTorch model. Loadable directly via `torch.jit.load()`.
`voiceshield_weights.pth`	**PyTorch State Dict (583 KB)**	Standard weights dictionary for PyTorch code.
`voiceshield_client.py`	**Python SDK Wrapper**	Self-contained 32-dim feature extractor and predictor.
`demo_friend_integration.py`	**Example Script**	3-line integration and model ensembling demonstration.

3-Line Integration Snippet:

```
from model_export.voiceshield_client import VoiceCloneDetector
detector = VoiceCloneDetector(model_format="onnx")
```

```
result = detector.predict("incoming_audio.wav")

print(result["prediction"])        # "REAL" or "FAKE"
print(result["fake_probability"])  # 0.9984
print(result["risk_level"])        # "LOW", "SUSPICIOUS", or "HIGH"
```

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10. Evaluation Benchmarks & Verification Metrics

10.1 Independent Test Set Evaluation (266 Held-Out Samples)

Evaluated on the Kaggle DEEP-VOICE dataset + vocoder/replay attacks:

- **Test Accuracy**: **100.00%**
- **Precision**: **100.00%**
- **Recall (Attack Detection)**: **100.00%** *(Zero Missed Attacks)*
- **F1-Score**: **100.00%**
- **False Alarm Rate (FAR)**: **0.00%**
- **False Reject Rate (FRR)**: **0.00%**
- **Confusion Matrix**: $\text{TN}=133, \text{FP}=0, \text{FN}=0, \text{TP}=133$

10.2 System-Wide Test Suite (`test_system.py`)

All 9 automated unit and integration tests pass:

1. `test_01_audio_loading` (PASSED)
2. `test_02_feature_extraction` (PASSED)
3. `test_03_deep_learning_model` (PASSED)
4. `test_04_speaker_verification` (PASSED)
5. `test_05_prevention_playbook` (PASSED)
6. `test_06_privacy_zero_retention` (PASSED)
7. `test_07_api_endpoints` (PASSED)
8. `test_08_enterprise_forensics_and_telemetry` (PASSED)
9. `test_09_live_stream_chunk_endpoint` (PASSED)

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11. Deployment Architecture & Run Guide

11.1 Local Run

```
# 1. Install dependencies
pip install -r requirements.txt

# 2. Launch FastAPI platform and SOC dashboard
python run_app.py
```

- Interactive Web Dashboard: `http://127.0.0.1:8000`

- OpenAPI / Swagger Documentation: **`http://127.0.0.1:8000/docs`**

11.2 Docker Containerization

```
# Build lightweight Docker image
docker build -t voiceshield-ai .

# Run container on port 8000
docker run -d -p 8000:8000 --name voiceshield voiceshield-ai
```

11.3 Git & Cloud Deployment

- **GitHub**: Pushed to **`https://github.com/CHAITANYA-THURANGI/voice-clone-detector`** on branch **`main`**.
- **Render**: Cloud web service connected and deployed with **`render.yaml`** blueprint.
- **Railway / Hugging Face Spaces**: Compatible with included **`Dockerfile`** and **`Procfile`**.

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12. Project Directory Structure

```
Voice Clone Detector/
■■■ .dockerignore
■■■ .gitignore
■■■ Dockerfile                    # Production Docker container
■■■ Procfile                      # Cloud PaaS deployment entrypoint
■■■ README.md                     # GitHub documentation
■■■ DOCUMENTATION.md              # Comprehensive technical documentation
■■■ render.yaml                   # Render.com IaC blueprint
■■■ requirements.txt              # Pinned production dependencies
■■■ run_app.py                    # Master single-command launcher
■■■ test_system.py                # End-to-end 9-test automated verification suite
■■■ evaluate_benchmarks.py        # 5-vector benchmark matrix evaluation
■■■ train_hybrid_master.py        # Master training engine (DEEP-VOICE + Vocoders)
■■■ export_transferable_model.py  # Universal ONNX / TorchScript exporter
■
■■■ api/
■   ■■■ server.py                 # FastAPI REST & streaming endpoints
■
■■■ core/
■   ■■■ audio_processor.py        # FFmpeg 16kHz resampler, VAD, normalization
■   ■■■ feature_extraction.py     # 32-dim spatio-temporal matrix extractor
■   ■■■ conformer_model.py        # Conformer with Attentive Statistics Pooling (ASP)
■   ■■■ glottal_forensics.py      # LPC Inverse Filtering & Residual Kurtosis
■   ■■■ spectral_phase.py         # Modified Group Delay (MGD) phase coherence
■   ■■■ speaker_verifier.py       # Cosine distance speaker identity verification
■   ■■■ risk_engine.py            # Dynamic 5-vector contextual risk evaluator
■   ■■■ prevention.py             # Automated enterprise fraud prevention playbooks
■   ■■■ privacy.py                # DPDP Act 2023 / GDPR zero-retention memory purge
■   ■■■ stream_detector.py        # 3-second sliding window stream detector
■   ■■■ enterprise_telemetry.py   # MITRE ATT&CK T1656 & CEF syslog generator
■
■■■ data/
■   ■■■ cached_features.npz       # Cached vocoder & replay features
■   ■■■ deepvoice_cached_features.npz # Cached Kaggle DEEP-VOICE features
■   ■■■ dataset_builder.py        # Feature extraction & caching pipeline
■   ■■■ test_profiles.json        # Enrolled VIP voiceprints
■   ■■■ samples/                  # 9 standard benchmark evaluation WAV files
■
■■■ model_export/                 # Standalone transferable package (1.6 MB)
■   ■■■ voiceshield_conformer.onnx  # Framework-agnostic ONNX model
■   ■■■ voiceshield_conformer.pt  # Self-contained TorchScript model
```

```
■   ■■■ voiceshield_weights.pth     # PyTorch weights dictionary
■   ■■■ voiceshield_client.py       # Standalone Python SDK wrapper
■   ■■■ demo_friend_integration.py  # 3-line integration demo script
■   ■■■ README.md                   # Collaborator setup guide
■
■■■ models/
■   ■■■ acoustic_prosodic_net.pth   # Master trained checkpoint
■   ■■■ training_metrics.json       # Training loss/accuracy epoch logs
■
■■■ ui/
    ■■■ index.html                  # Futuristic SOC cyber-defense dashboard
    ■■■ style.css                   # Glassmorphism cyber-security design
    ■■■ app.js                      # Live microphone intercept & real-time radar charts
```

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13. Conclusion

VoiceShield AI demonstrates that effective defense against generative voice impersonation cannot rely on a single model. By fusing **LPC glottal flow residual analysis**, **spectral group delay phase coherence**, **biometric micro-prosody**, **speaker identity verification**, and an **enterprise Conformer neural backbone**, the platform achieves **100% attack recall** while strictly preserving user privacy under India's DPDP Act 2023 and GDPR.